

Better resource allocation and network traffic management. MCFs facilitate efficient resource allocation and better management of larger traffic, as each core can be dedicated to specific applications in an SDM system

Enhanced redundancy and reliability. An MCF offers built-in redundancy. If a specific core fails, another core can be used to maintain operation, thereby improving network reliability

Multiband optical network

Adopting MCF in place of traditional SCF addresses bandwidth challenges, but it is a capital-intensive (CAPEX) solution. Another approach is using multiband optical networks. Currently, most optical networks use the C-band (1530nm-1565nm) for data transmission, with some extending to the L-band (1565nm-1625nm) to increase bandwidth.

Multiband optical networks use multiple wavelength bands for signal transmission via WDM in standard optical fibre (Fig. 5). Utilising multiple optical bands enables these networks to achieve an overall bandwidth over ten times greater than conventional systems that only use the C-band.

Multiband optical networks offer increased data capacity, improved spectral efficiency, and enhanced network resiliency compared to single-band networks. However, challenges include the need for new multiband amplification techniques, new multiband transceivers, advanced control plane solutions, and efficient network planning and control.

Standardisation activities

Multicore fibre and multiband optical networks are emerging technologies with ongoing research and development, so standardisation activities remain limited. Nevertheless, ITU-T has proposed standards for initial SDM development.

According to the ITU-T technical report, GSTR-SDM – Optical fibre, cable, and components for space division multiplexing transmission:

For SDM optical fibre links, the following entities must be harmonised for a physical layer standard:

1. [IEC SC 86B]
2. [IEC SC 86C]
3. ITU-T Q6/SG15
4. ITU-T Q2/SG15

Achieving very low crosstalk with a relatively simple refractive index profile while maintaining a cladding diameter of 125µm may be difficult. However, optical compatibility with existing ITU-T G.652, ITU-T G.654, and ITU-T G.657 fibres is essential.



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